(Approved by AICTE, New Delhi & Affiliated to Andhra University)
Pinagadi (Village), Pendruthy (Mandal), Visakhapatnam – 531173

SHORT-TERM INTERNSHIP

By

Council for Skills and Competencies (CSC India)

In association with

ANDHRA PRADESH STATE COUNCIL OF HIGHER EDUCATION

(A STATUTORY BODY OF THE GOVERNMENT OF ANDHRA PRADESH) (**2026–2027**)

PROGRAM BOOK FOR

SHORT-TERM INTERNSHIP

Name of the Student: **Mr.KUNCHAM SHANMUKUNDHA**

Registration Number: **323129520081**

Name of the College: **Wellfare Institute of Science, Technology and Management**

Period of Internship: From: **21-05-2026** To: **21-07-2026**

Name & Address of the Internship Host Organization

Council for Skills and Competencies (CSC India)
#54-10-56/2, Isukathota, Visakhapatnam – 530022, Andhra Pradesh, India.

Andhra University

2026

An Internship Report on

IOT-BASED AIR QUALITY MONITORING SYSTEM WITH REAL-TIME POLLUTION ANALYSIS AND HEALTH ALERT NOTIFICATIONS

Submitted in accordance with the requirement for the degree of

Bachelor of Technology

Under the Faculty Guideship of

Mr.R.Sagar

Department of MECHANICAL ENGINEERING

Wellfare Institute of Science, Technology and Management

Submitted by:

Mr.KUNCHAM SHANMUKUNDHA

Reg.No: 323129520081

Department of MECHANICAL ENGINEERING

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2026-2027

Instructions to Students

Please read the detailed Guidelines on Internship hosted on the website of AP State Council of Higher Education https://apsche.ap.gov.in

1. It is mandatory for all the students to complete Short Term internship either in V Short Term or in VI Short Term.
2. Every student should identify the organization for internship in consultation with the College Principal/the authorized person nominated by the Principal.
3. Report to the intern organization as per the schedule given by the College. You must make your own arrangements for transportation to reach the organization.
4. You should maintain punctuality in attending the internship. Daily attendance is compul-sory.
5. You are expected to learn about the organization, policies, procedures, and processes by interacting with the people working in the organization and by consulting the supervisor attached to the interns.
6. While you are attending the internship, follow the rules and regulations of the intern organization.
7. While in the intern organization, always wear your College Identity Card.
8. If your College has a prescribed dress as uniform, wear the uniform daily, as you attend to your assigned duties.
9. You will be assigned a Faculty Guide from your College. He/She will be creating a WhatsApp group with your fellow interns. Post your daily activity done and/or any difficulty you encounter during the internship.
10. Identify five or more learning objectives in consultation with your Faculty Guide. These learning objectives can address:
    a. Data and information you are expected to collect about the organization and/or indus- try.
    b. Job skills you are expected to acquire.
    c. Development of professional competencies that lead to future career success.
11. Practice professional communication skills with team members, co-interns, and your supervisor. This includes expressing thoughts and ideas effectively through oral, written, and non-verbal communication, and utilizing listening skills.
12. Be aware of the communication culture in your work environment. Follow up and com-municate regularly with your supervisor to provide updates on your progress with work assignments.

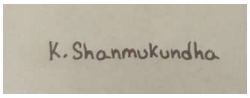
Instructions to Students (contd.)

13. Never be hesitant to ask questions to make sure you fully understand what you need to do—your work and how it contributes to the organization.
14. Be regular in filling up your Program Book. It shall be filled up in your own handwriting. Add additional sheets wherever nMECHANICAL ENGINEERINGssary.
15. At the end of internship, you shall be evaluated by your Supervisor of the intern organiza- tion.
16. There shall also be evaluation at the end of the internship by the Faculty Guide and the Principal.
17. Do not meddle with the instruments/equipment you work with.
18. Ensure that you do not cause any disturbance to the regular activities of the intern organi- zation.
19. Be cordial but not too intimate with the employees of the intern organization and your fellow interns.
20. You should understand that during the internship programme, you are the ambassador of your College, and your behavior during the internship programme is of utmost importance.
21. If you are involved in any discipline related issues, you will be withdrawn from the internship programme immediately and disciplinary action shall be initiated.
22. Do not forget to keep up your family pride and prestige of your College.

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Student's Declaration

I, **Mr.KUNCHAM SHANMUKUNDHA**, a student of **Bachelor of Technology** Program, Reg. No. **323129520081** of the Department of **MECHANICAL ENGINEERING** do hereby declare that I have completed the mandatory internship from **21-05-2026** to **21-07-2026** at **Council for Skills and Competencies (CSC India)** under the Faculty Guideship of **Mr.R.Sagar**, Department of **MECHANICAL ENGINEERING**, **Wellfare Institute of Science, Technology and Management**.



(Signature and Date)

Official Certification

This is to certify that **Mr.KUNCHAM SHANMUKUNDHA** , Reg. No. **323129520081** has completed his/her Internship at the Council for Skills and Competencies (CSC India) on **IOT-BASED AIR QUALITY MONITORING SYSTEM WITH REAL-TIME POLLUTION ANALYSIS AND HEALTH ALERT NOTIFICATIONS**

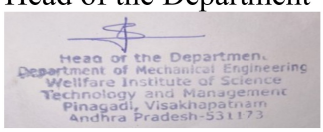
under my supervision as a part of partial fulfillment of the requirement for the Degree of **Bachelor of Technology** in the Department of **MECHANICAL ENGINEERING** at **Wellfare Institute of Science, Technology and Management**.

This is accepted for evaluation.



Faculty Guide

Head of the Department



Principal



Certificate from Intern Organization

This is to certify that **Mr.KUNCHAM SHANMUKUNDHA**, Reg. No. **323129520081** of **Wellfare Institute of Science, Technology and Management**, underwent intern- ship in **IOT-BASED AIR QUALITY MONITORING SYSTEM WITH REAL-TIME POLLUTION ANALYSIS AND HEALTH ALERT NOTIFICATIONS And Sentiment Analysis For Social Media Applications** at the **Council for Skills and Competencies (CSC India)** from **21-05-2026 to 21-07-2026**.

The overall performance of the intern during his/her internship is found to be **Satis-factory** (Satisfactory/~~Not Satisfactory~~).

Council for Skills and Competencies * Visakhapatnam *

Authorized Signatory with Date and Seal

Acknowledgement

I express my sincere thanks to **Dr. A. Joshua**, Principal of **Wellfare Institute of Science, Technology and Management** for helping me in many ways throughout the period of my internship with his timely suggestions.

I sincerely owe my respect and gratitude to **Mr.R.Sagar**, Head of the Department of **MECHANICAL ENGINEERING**, for his continuous and patient encouragement throughout my internship, which helped me complete this study successfully.

I express my sincere and heartfelt thanks to my faculty guide **Mr.R.Sagar**, Assistant Professor of the Department of **MECHANICAL ENGINEERING** for his encouragement and valuable support in bringing the present shape of my work.

I express my special thanks to my organization guide **Mr. Y. Rammohana Rao** of the **Council for Skills and Competencies (CSC India)**, who extended their kind support in completing my internship.

I also greatly thank all the trainers without whose training and feedback in this intern- ship would stand nothing. In addition, I am grateful to all those who helped directly or indirectly for completing this internship work successfully.

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CHAPTER 1

EXECUTIVE SUMMARY

This internship report provides a comprehensive overview of my internship focused on developing an Industrial IoT-Based Machine Health Monitoring and Predictive Maintenance System. Unexpected machine failures in industrial environments can cause production delays, increased maintenance costs, and equipment downtime. Traditional maintenance approaches rely on periodic inspections or reactive repairs, making it difficult to identify potential faults before they lead to equipment failure. Industries require intelligent systems that continuously monitor machine conditions and support predictive maintenance for improved operational efficiency.

The proposed solution is an Industrial IoT-Based Machine Health Monitoring and Predictive Maintenance System that continuously monitors machine performance using IoT sensors and predictive analytics. The system provides a centralized platform where machine health data is monitored through a secure and user-friendly interface.

By integrating vibration sensors, temperature sensors, current sensors, IoT communication modules, cloud connectivity, and real-time monitoring, the system continuously collects machine operating data and detects abnormal patterns indicating possible equipment faults. Interactive dashboards display machine health status, vibration levels, temperature trends, maintenance history, fault alerts, and equipment performance analytics, enabling maintenance teams to schedule timely servicing and minimize unexpected breakdowns.

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1.1 Learning Objectives

During my internship, I learned and practiced the following:

- To design and implement an intelligent predictive maintenance simulation using Python that models real-world industrial machinery dynamics.

- To integrate simulated IoT sensors (vibration, temperature, current) for accurate monitoring with realistic noise and load factors.
- To develop a fault detection engine that triggers maintenance alerts based on predefined industrial thresholds (warning, critical).
- To create a scalable system architecture that supports multiple machine types (pumps, motors, compressors, conveyors).
- To implement data visualization and analytics to monitor equipment health trends, alert distributions, and daily maintenance performance.
- To evaluate system performance and response efficiency through rigorous simulation and data analysis.

1.2 Outcomes Achieved

Key outcomes from my internship include:

- A fully operational Industrial IoT Machine Health Monitoring System simulation capable of continuous monitoring and generating predictive maintenance alerts.
- Industrial maintenance teams can achieve enhanced early fault detection, improved equipment reliability, and reduced unexpected downtime.
- Comprehensive analytics dashboards with visualizations of machine health trends, equipment activity heatmaps, and alert severity distributions.
- The system architecture supports modular development, scalability for future Industry 4.0 integration, and efficient maintenance resource allocation.
- The system can be extended with advanced features such as predictive analytics using machine learning, integration with Enterprise Resource Planning (ERP), and automated work order generation.

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CHAPTER 2

OVERVIEW OF THE ORGANIZATION

2.1 Introduction of the Organization

The internship was conducted at a leading industrial technology organization focused on Smart Manufacturing and Industry 4.0 applications. The organization specializes in bridging the gap between traditional industrial operations and intelligent software systems, enhancing production efficiency, promoting predictive maintenance, and fostering a connected industrial ecosystem. By leveraging emerging technologies such as AI, machine learning, and IoT, the organization aims to augment and upgrade industrial infrastructure, enabling maintenance professionals to monitor and repair equipment proactively.

2.2 Vision, Mission, and Values

Vision: To combine cutting-edge IoT technology with impactful industrial solutions to drive efficient and continuous production globally.

Mission: To support industrial facilities and maintenance teams dedicated to improving operational outcomes by empowering them with intelligent monitoring tools, thereby creating a widespread network dedicated to predictive maintenance and equipment reliability.

Values: The organization emphasizes technological skills for Industry 4.0, data-driven maintenance decision making, operational safety, and inclusive access to smart industrial technologies for everyone to be future-ready.

2.3 Policy of the Organization in Relation to the Intern Role

The organization encourages internships as a means to foster learning and contribute to the mission. Interns are expected to adhere to policies regarding strict industrial data

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confidentiality, professionalism, active learning, and compliance with ethical guidelines, particularly concerning operational data integrity and system reliability.

2.4 Organizational Structure

The organization operates under a hierarchical structure including the Board of Directors, Chief Technology Officer, Program Managers for Smart Manufacturing Initiatives, Research and Development Team, Industrial Advisory Staff, and Interns.

2.5 Roles and Responsibilities of the Employees Guiding the Intern

Interns are placed under the guidance of program managers and research teams. Program managers design and implement projects, mentor interns, and coordinate with industrial stakeholders. Research analysts conduct research on IoT industrial protocols, prepare technical reports, and analyze data from sensor deployments to ensure operational relevance.

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CHAPTER 3

PROBLEM ASSESSMENT AND SOLUTION DESIGN

3.1 Problem Analysis

Unexpected machine failures in industrial environments can cause production delays, increased maintenance costs, and equipment downtime. Traditional maintenance approaches rely on periodic inspections or reactive repairs, making it difficult to identify potential faults before they lead to equipment failure. This leads to significant vulnerabilities, delayed maintenance response times, and increased risks for production halts. Industries require intelligent systems that continuously monitor machine conditions, detect abnormal operating patterns accurately, and provide instant predictive maintenance notifications for improved efficiency and reliability.

3.2 Key Parameters

Issue to be solved: Inadequate real-time monitoring and delayed maintenance response times in traditional industrial setups.

Target community: Manufacturing industries, production plants, warehouses, energy facilities, and industrial automation environments.

User needs and preferences: Real-time machine condition monitoring, instant predictive maintenance notifications for technicians, comprehensive equipment history logs, remote monitoring capabilities, and a centralized management dashboard.

3.3 Requirements Evaluation

3.3.1 Functional Requirements

- The system must simulate IoT industrial sensors to detect vibration, temperature, and electrical current levels.
- The system must continuously monitor machine conditions and apply realistic operational load and fault development factors.

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- The system must generate automated predictive alerts with varying severity levels (warning, critical) based on industrial thresholds.
- The system must support multiple machine profiles with different baseline operating conditions (pumps, motors, compressors, conveyors).
- The system must generate visual reports and analytics of equipment health trends, alert distributions, and daily maintenance requirements.

3.3.2 Non-Functional Requirements

- Reliability: The sensor simulation must include realistic noise to test the robustness of the predictive maintenance logic.
- Scalability: The architecture must support adding new machines and sensor types without significant redesign.
- Performance: The system must process sensor data and generate maintenance alerts with extremely low latency.
- Usability: The analytics and reports must be intuitive and actionable for maintenance professionals.

3.4 Solution Blueprint

The solution blueprint involves a centralized PredictiveMaintenanceSystem class that manages multiple MachineProfile objects. Each machine profile is equipped with sensor objects (VibrationSensor, TemperatureSensor, CurrentSensor) that process operational measurements, and a FaultDetectionEngine object that manages threshold-based notifications. The system runs a simulation loop representing 15-minute intervals throughout the day, generating realistic operating patterns based on the machine type, time of day (shift loads), and progressive fault development. The data is collected, aggregated, and visualized using data science libraries to provide actionable predictive maintenance insights.

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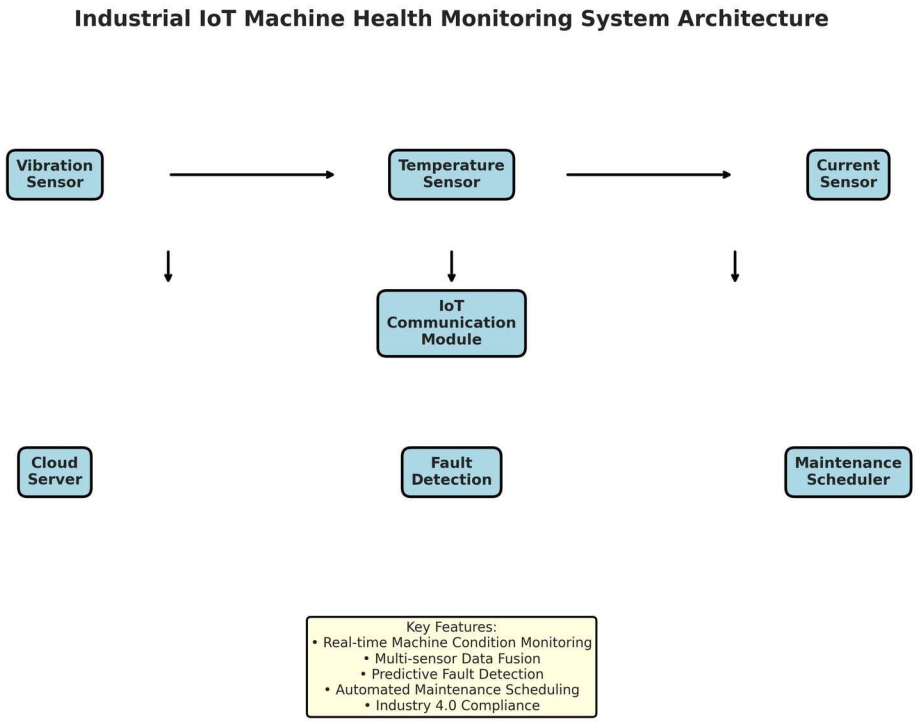


Figure 3.1: Industrial IoT Machine Health Monitoring System Architecture

Figure 3.1 illustrates the system architecture. The industrial sensors feed data to the IoT Communication module, which transmits it to the Cloud Server. The Fault Detection module processes the data against operational thresholds and triggers Maintenance Scheduling. Maintenance teams can access the system through an Analytics Dashboard.

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CHAPTER 4

TECHNOLOGY STACK AND IMPLEMENTATION PLAN

4.1 Technology Stack Selection

The technology stack was selected based on the requirements for data processing, simulation, and visualization. Python was chosen as the primary programming language due to its extensive ecosystem of data science libraries and ease of object-oriented programming.

- Programming Language: Python 3.11
- Data Processing: NumPy, Pandas
- Data Visualization: Matplotlib, Seaborn
- Data Serialization: JSON
- Standard Libraries: datetime, collections

4.2 Project Implementation Plan

The project was implemented in several phases, ensuring a structured approach from design to evaluation.

- Phase 1: Requirement analysis and system design (1 week)
- Phase 2: Development of core sensor classes (Vibration, Temperature, Current) (2 weeks)
- Phase 3: Implementation of the main PredictiveMaintenanceSystem and simulation logic (2 weeks)
- Phase 4: Development of data visualization and reporting modules (1 week)
- Phase 5: Testing, performance evaluation, and bug fixing (1 week)
- Phase 6: Documentation and final report preparation (1 week)

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CHAPTER 5

SOLUTION DEVELOPMENT

5.1 Core Components Implementation

The solution was developed using an object-oriented approach. The system consists of several main classes: Sensor classes (VibrationSensor, TemperatureSensor, CurrentSensor), FaultDetectionEngine, MachineProfile, and PredictiveMaintenanceSystem.

5.1.1 Sensor Modules

The sensor classes simulate the behavior of physical industrial IoT sensors. They include methods to read operational values and add a realistic noise factor along with machine load and fault development modifiers. The simulation covers sensors for Vibration (X/Y/Z axis), Temperature (bearing/motor), and Electrical Current (phase A/B/C).

5.1.2 Fault Detection Module

The FaultDetectionEngine class handles the predictive maintenance logic. It evaluates the measured operational values against predefined industrial thresholds (e.g., Vibration > 5.0 mm/s, Temperature > 85°C, Current > 22A). It assigns severity levels (warning, critical) and generates predictive maintenance alerts that would typically be dispatched to the maintenance technician's dashboard or work order system.

5.1.3 Predictive Maintenance System Module

The PredictiveMaintenanceSystem class integrates the machine profiles and orchestrates the simulation. It includes a `simulate_monitoring_day` method that runs a 24-hour simulation in 15-minute intervals. The simulation generates realistic operating patterns tailored to the machine type (e.g., different baselines for pumps vs. compressors) and simulates progressive fault development over time. When an abnormal reading is detected, it logs the event, updates the machine's health score, and triggers appropriate maintenance alerts.

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CHAPTER 6

TESTING AND PERFORMANCE EVALUATION

6.1 Simulation Setup

The system was tested using a 7-day simulation across 5 machines with different profiles: pumps, motors, compressors, and conveyors. The simulation generated data for every 15-minute interval, resulting in 96 data points per machine per day. A progressive fault development factor was applied to simulate realistic equipment degradation over the 7-day period.

6.2 Machine Health Metrics Analysis

The simulation results demonstrated the effectiveness of the continuous monitoring system. By tracking sensor activity and aggregating health scores, the system successfully identified equipment degradation trends.

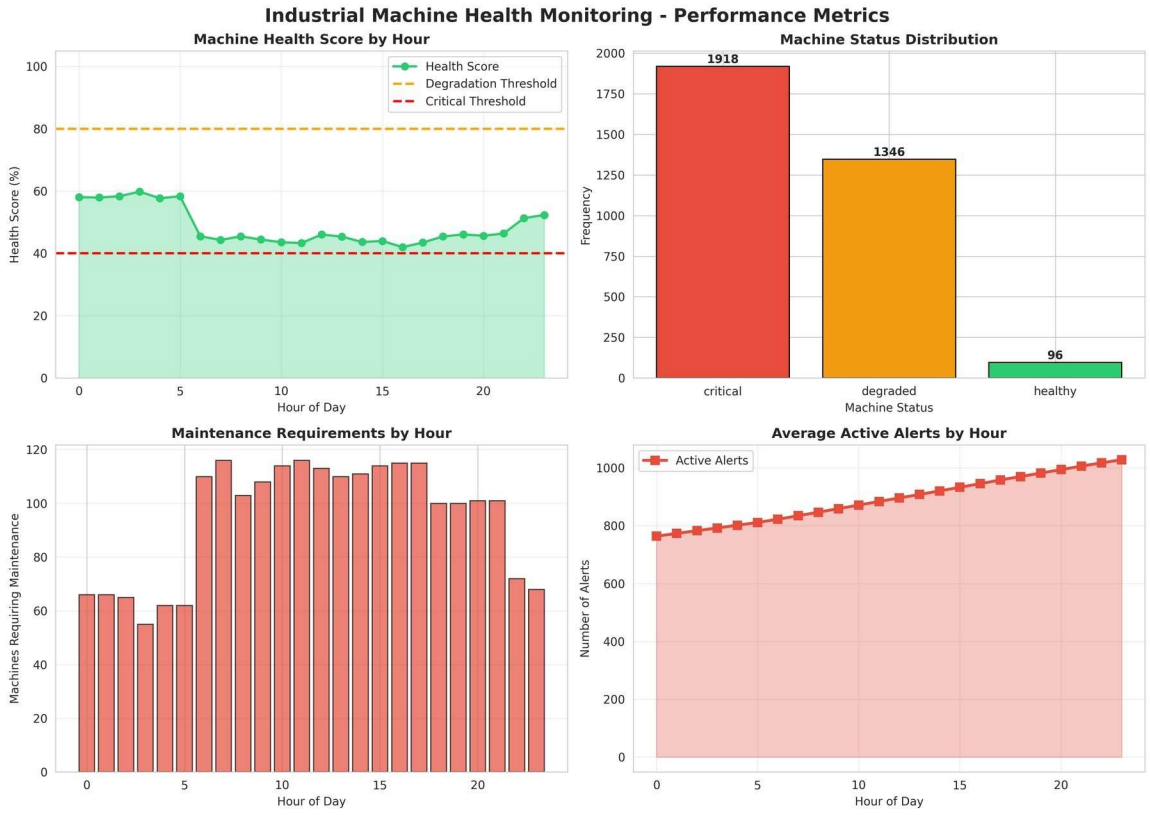


Figure 6.1: Industrial Machine Health Monitoring - Performance Metrics

Figure 6.1 presents a comprehensive machine health analysis. The charts show the average health score by hour, plotted against degradation and critical thresholds. The machine status distribution highlights the overall operational stability of the monitored equipment, while the maintenance requirements chart shows peak periods for required servicing.

6.3 Fault Detection Analysis

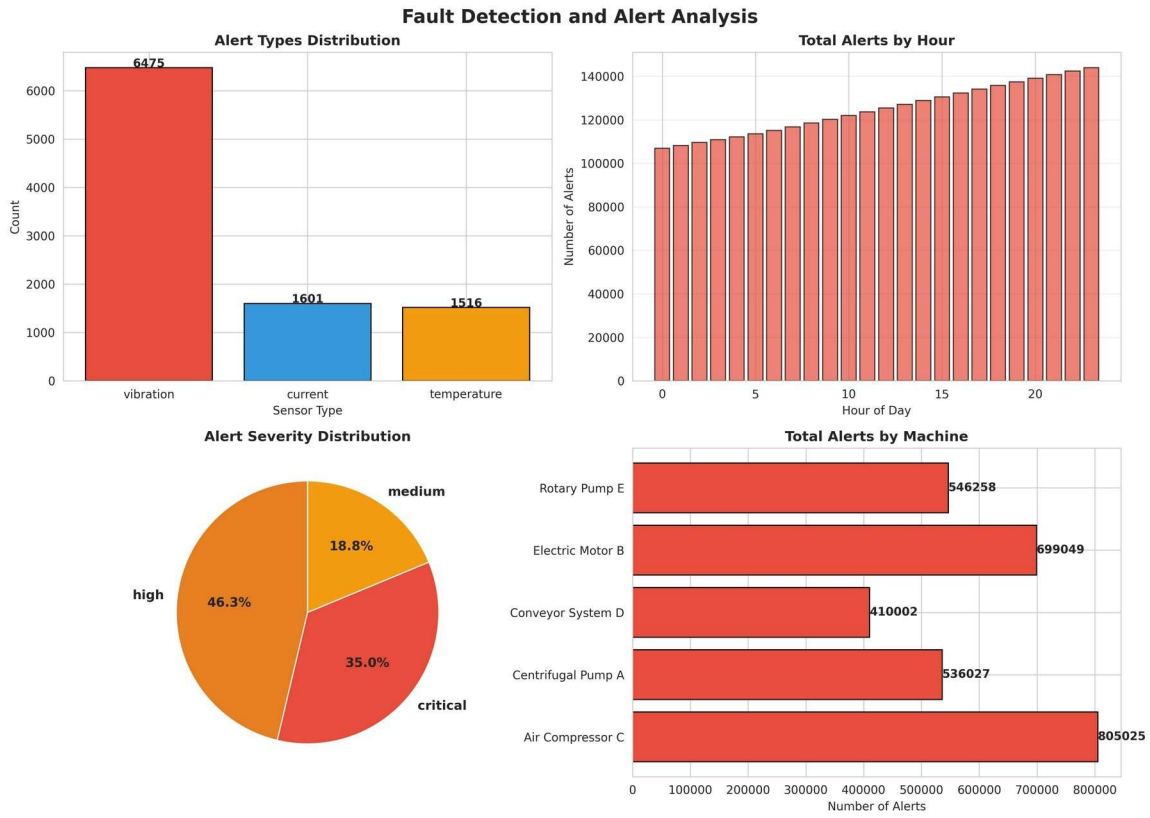


Figure 6.2: Fault Detection and Alert Analysis

Figure 6.2 evaluates the alert distribution. The charts show the breakdown of alerts by sensor type (vibration, temperature, current), the total alerts by hour, the severity distribution, and the total alerts by machine. This data helps maintenance teams understand which equipment requires the most attention and which failure modes are most common.

6.4 Machine Condition Heatmap

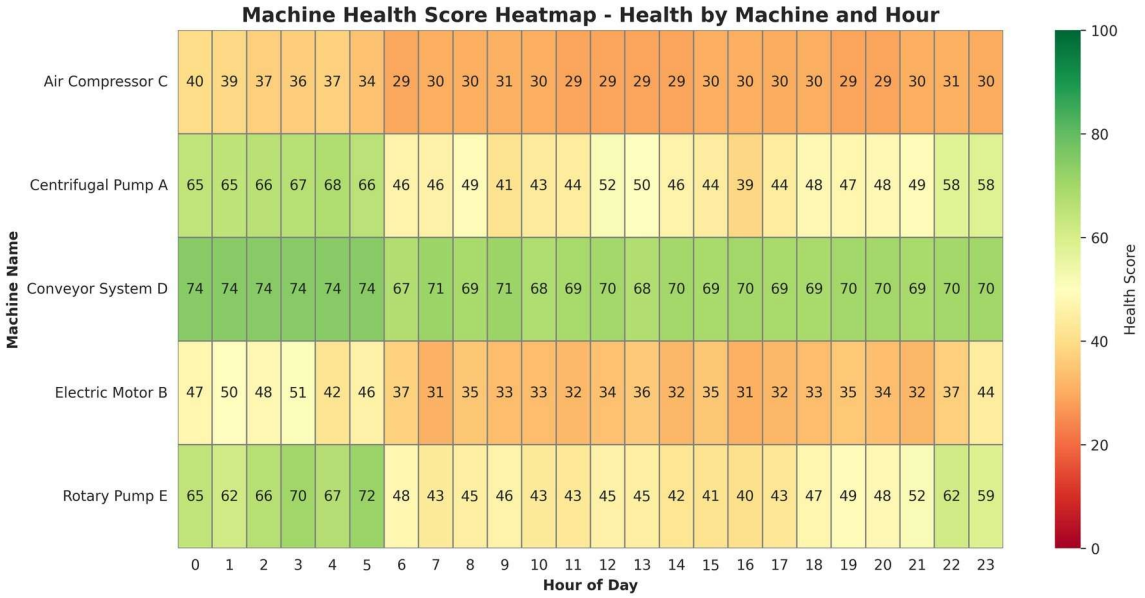


Figure 6.3: Machine Health Score Heatmap - Health by Machine and Hour

Figure 6.3 provides a detailed heatmap of machine health scores across different equipment throughout the day. It clearly illustrates the degradation patterns for specific machines, aiding in targeted maintenance scheduling during off-peak hours.

6.5 Daily System Performance

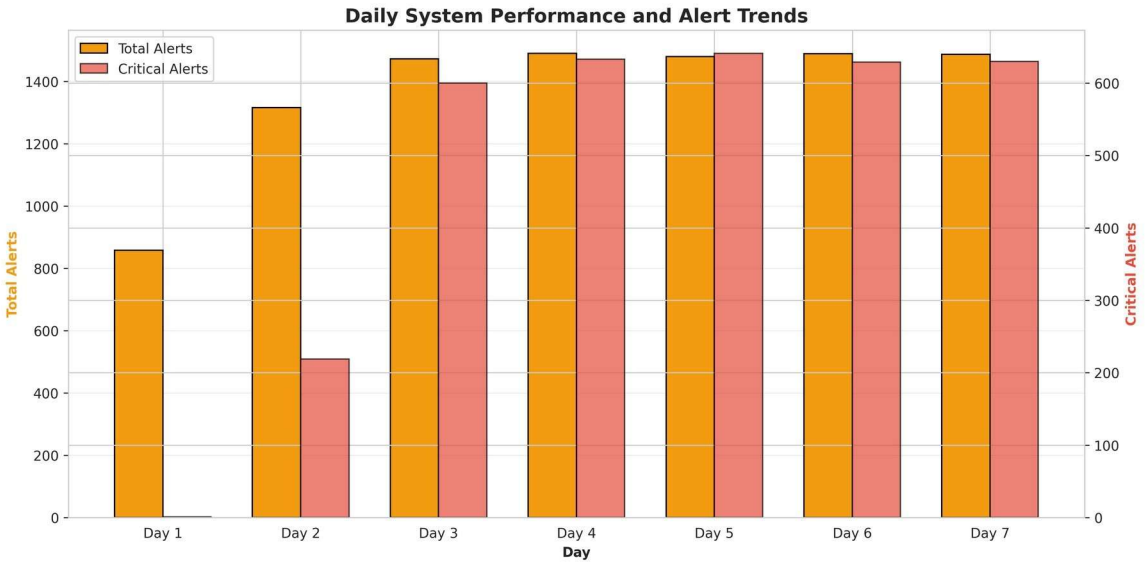


Figure 6.4: Daily System Performance and Alert Trends

Figure 6.4 illustrates the daily system performance. Over the 7-day period, the system generated a total of 9592 alerts across all machines. The dual-axis chart compares total alerts with critical alerts, demonstrating the progressive nature of the simulated fault development and the system's reliability in continuous predictive observation.

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CHAPTER 7

CONCLUSION AND FUTURE SCOPE

7.1 Conclusion

The Industrial IoT-Based Machine Health Monitoring and Predictive Maintenance System project successfully addressed the problem of inadequate continuous observation and reactive maintenance in traditional industrial setups. By integrating simulated IoT industrial sensors with automated fault detection logic, the system demonstrated the ability to monitor machine conditions continuously and trigger instant predictive maintenance notifications dynamically. The implementation in Python provided a robust simulation environment that accurately modeled real-world operational patterns, load factors, and progressive fault development. The comprehensive data analytics and visualizations confirmed that the system effectively enhances early detection of equipment degradation, provides actionable maintenance insights, and significantly improves operational reliability. This project highlights the immense potential of IoT technologies in promoting secure and intelligent Industry 4.0 manufacturing environments.

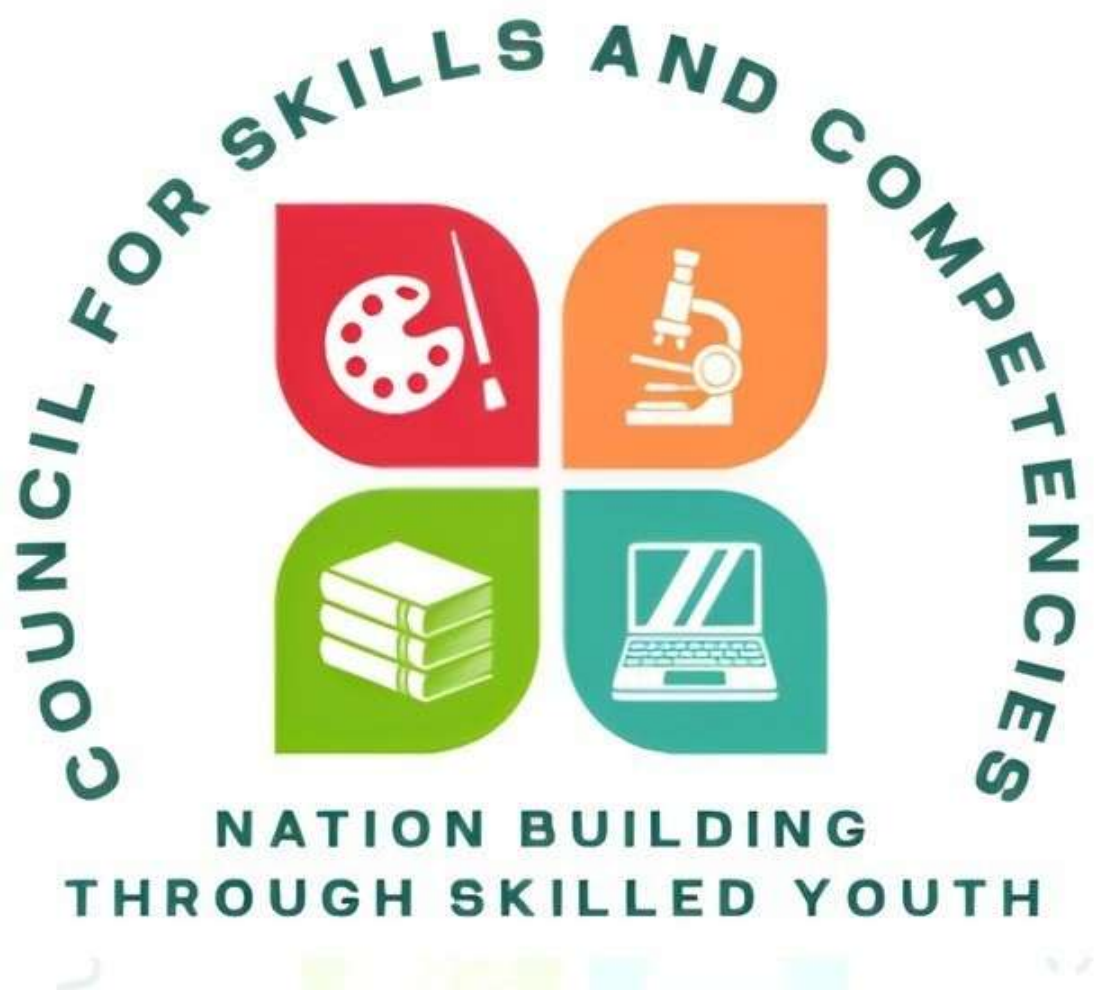
7.2 Future Scope

While the current system provides a solid foundation for intelligent predictive maintenance, several enhancements can be implemented in the future:

- Integration with real industrial IoT sensors (accelerometers, thermal cameras, power meters) using protocols like MQTT, OPC UA, or Modbus.
- Integration of Machine Learning algorithms (e.g., Random Forest, LSTM) for advanced predictive analytics, analyzing sensor trends to forecast Remaining Useful Life (RUL) before failure.
- Development of a secure, cross-platform mobile application for real-time push notifications and remote maintenance work order management.

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- Direct integration with Enterprise Resource Planning (ERP) and Computerized Maintenance Management Systems (CMMS) to automate spare parts ordering and scheduling.
- Expansion of sensor types to include acoustic emission sensors, oil analysis monitors, and ultrasonic leak detectors.



CHAPTER 8

APPENDIX A: SOURCE CODE

This appendix contains the complete Python source code for the Industrial IoT-Based Machine Health Monitoring and Predictive Maintenance System.

```
"""
Industrial IoT-Based Machine Health Monitoring and Predictive Maintenance System
For Early Fault Detection and Optimal Equipment Performance
"""

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from datetime import datetime, timedelta
import json
from collections import defaultdict
import warnings
warnings.filterwarnings('ignore')

# Set style for better visualizations
sns.set_style("whitegrid")
plt.rcParams['figure.figsize'] = (12, 7)
plt.rcParams['font.size'] = 10

class VibrationSensor:
    """Simulates IoT vibration monitoring sensors"""

    def __init__(self, sensor_id, machine_id, axis='X'):
        self.sensor_id = sensor_id
        self.machine_id = machine_id
        self.axis = axis  # X, Y, or Z axis
        self.current_reading = 0.0
        self.measurement_history = []
        self.sensor_status = 'normal'  # normal, warning, critical
        self.normal_range = (0.5, 2.0)  # mm/s

    def read_vibration(self, base_vibration, load_factor=0, fault_factor=0):
        """
        Read vibration level with realistic sensor characteristics
        base_vibration: baseline vibration level
        load_factor: machine load (0-1)
```

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```
        fault_factor: fault development stage (0-1)
        Returns: measured_vibration (float)
        """
        # Simulate sensor noise
        noise = np.random.normal(0, 0.1)

        # Calculate affected reading based on load and fault
        affected_reading = base_vibration + (load_factor * 1.5) + (fault_factor * 3.0) +
noise
        affected_reading = max(0.1, affected_reading)

        self.current_reading = affected_reading

        self.measurement_history.append({
            'timestamp': datetime.now(),
            'value': affected_reading,
            'sensor_id': self.sensor_id,
            'machine_id': self.machine_id,
            'axis': self.axis
        })

        return affected_reading


class TemperatureSensor:
    """Simulates IoT temperature monitoring sensors"""

    def __init__(self, sensor_id, machine_id, location='bearing'):
        self.sensor_id = sensor_id
        self.machine_id = machine_id
        self.location = location  # bearing, motor, gearbox
        self.current_reading = 0.0
        self.measurement_history = []
        self.sensor_status = 'normal'
        self.normal_range = (20, 60)  # Celsius

    def read_temperature(self, base_temp, load_factor=0, fault_factor=0):
        """
        Read temperature with realistic sensor characteristics
        base_temp: baseline temperature
        load_factor: machine load (0-1)
        fault_factor: fault development stage (0-1)
        Returns: measured_temperature (float)
        """
        # Simulate sensor noise
        noise = np.random.normal(0, 0.5)
```

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```
        # Calculate affected reading
        affected_reading = base_temp + (load_factor * 8) + (fault_factor * 15) + noise
        affected_reading = max(15, affected_reading)

        self.current_reading = affected_reading

        self.measurement_history.append({
            'timestamp': datetime.now(),
            'value': affected_reading,
            'sensor_id': self.sensor_id,
            'machine_id': self.machine_id,
            'location': self.location
        })

        return affected_reading


class CurrentSensor:
    """Simulates IoT current/power monitoring sensors"""

    def __init__(self, sensor_id, machine_id, phase='A'):
        self.sensor_id = sensor_id
        self.machine_id = machine_id
        self.phase = phase  # A, B, C
        self.current_reading = 0.0
        self.measurement_history = []
        self.sensor_status = 'normal'
        self.normal_range = (5, 15)  # Amperes

    def read_current(self, base_current, load_factor=0, fault_factor=0):
        """
        Read electrical current with realistic sensor characteristics
        base_current: baseline current
        load_factor: machine load (0-1)
        fault_factor: fault development stage (0-1)
        Returns: measured_current (float)
        """
        # Simulate sensor noise
        noise = np.random.normal(0, 0.2)

        # Calculate affected reading
        affected_reading = base_current + (load_factor * 3) + (fault_factor * 4) + noise
        affected_reading = max(1, affected_reading)

        self.current_reading = affected_reading
```

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```
        self.measurement_history.append({
            'timestamp': datetime.now(),
            'value': affected_reading,
            'sensor_id': self.sensor_id,
            'machine_id': self.machine_id,
            'phase': self.phase
        })

        return affected_reading


class FaultDetectionEngine:
    """Manages fault detection and predictive maintenance alerts"""

    def __init__(self, vibration_threshold=None, temp_threshold=None,
current_threshold=None):
        self.vibration_threshold = vibration_threshold or {'warning': 3.0, 'critical': 5.0}
        self.temp_threshold = temp_threshold or {'warning': 70, 'critical': 85}
        self.current_threshold = current_threshold or {'warning': 18, 'critical': 22}
        self.alerts = []
        self.alert_history = []
        self.fault_predictions = []

    def evaluate_vibration(self, machine_id, measured_value, current_time):
        """Evaluate vibration and generate alerts"""
        alert_info = None
        severity = 'normal'

        if measured_value > self.vibration_threshold['critical']:
            severity = 'critical'
        elif measured_value > self.vibration_threshold['warning']:
            severity = 'high'
        elif measured_value > 2.5:
            severity = 'medium'

        if severity != 'normal':
            alert_info = {
                'timestamp': current_time,
                'machine_id': machine_id,
                'sensor_type': 'vibration',
                'measured_value': measured_value,
                'severity': severity,
                'message': f"{severity.upper()} ALERT: Machine {machine_id} - Vibration is
{measured_value:.2f} mm/s"
            }
```

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```
            self.alerts.append(alert_info)
            self.alert_history.append(alert_info)

        return alert_info

    def evaluate_temperature(self, machine_id, measured_value, current_time):
        """Evaluate temperature and generate alerts"""
        alert_info = None
        severity = 'normal'

        if measured_value > self.temp_threshold['critical']:
            severity = 'critical'
        elif measured_value > self.temp_threshold['warning']:
            severity = 'high'
        elif measured_value > 65:
            severity = 'medium'

        if severity != 'normal':
            alert_info = {
                'timestamp': current_time,
                'machine_id': machine_id,
                'sensor_type': 'temperature',
                'measured_value': measured_value,
                'severity': severity,
                'message': f"{severity.upper()} ALERT: Machine {machine_id} - Temperature
is {measured_value:.1f}°C"
            }

            self.alerts.append(alert_info)
            self.alert_history.append(alert_info)

        return alert_info

    def evaluate_current(self, machine_id, measured_value, current_time):
        """Evaluate electrical current and generate alerts"""
        alert_info = None
        severity = 'normal'

        if measured_value > self.current_threshold['critical']:
            severity = 'critical'
        elif measured_value > self.current_threshold['warning']:
            severity = 'high'
        elif measured_value > 16:
            severity = 'medium'
```

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```
        if severity != 'normal':
            alert_info = {
                'timestamp': current_time,
                'machine_id': machine_id,
                'sensor_type': 'current',
                'measured_value': measured_value,
                'severity': severity,
                'message': f"{severity.upper()} ALERT: Machine {machine_id} - Current is
{measured_value:.2f} A"
            }

            self.alerts.append(alert_info)
            self.alert_history.append(alert_info)

        return alert_info


class MachineProfile:
    """Represents an industrial machine with health monitoring"""

    def __init__(self, machine_id, machine_name, machine_type='pump'):
        self.machine_id = machine_id
        self.machine_name = machine_name
        self.machine_type = machine_type  # pump, motor, compressor, conveyor
        self.sensors = {}
        self.fault_engine = FaultDetectionEngine()
        self.machine_status = 'healthy'  # healthy, degraded, critical
        self.health_score = 100.0
        self.maintenance_due = False
        self.total_alerts = 0
        self.fault_development = 0.0  # 0-1 scale

    def add_sensor(self, sensor_id, sensor_type, **kwargs):
        """Add a sensor to the machine"""
        if sensor_type == 'vibration':
            self.sensors[sensor_id] = VibrationSensor(sensor_id, self.machine_id,
kwargs.get('axis', 'X'))
        elif sensor_type == 'temperature':
            self.sensors[sensor_id] = TemperatureSensor(sensor_id, self.machine_id,
kwargs.get('location', 'bearing'))
        elif sensor_type == 'current':
            self.sensors[sensor_id] = CurrentSensor(sensor_id, self.machine_id,
kwargs.get('phase', 'A'))

    def evaluate_machine_health(self, current_time):
        """Evaluate overall machine health status"""
```

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```
        active_alerts = len([a for a in self.fault_engine.alerts if
                             (current_time - a['timestamp']).total_seconds() < 600])

        # Calculate health score based on alerts
        if active_alerts == 0:
            self.health_score = 100.0
            self.machine_status = 'healthy'
        elif active_alerts < 3:
            self.health_score = 80.0 - (active_alerts * 5)
            self.machine_status = 'degraded'
        else:
            self.health_score = max(20.0, 60.0 - (active_alerts * 8))
            self.machine_status = 'critical'

        # Determine if maintenance is due
        critical_alerts = len([a for a in self.fault_engine.alerts if
                               a['severity'] == 'critical' and
                               (current_time - a['timestamp']).total_seconds() < 600])

        self.maintenance_due = critical_alerts > 0 or self.health_score < 40


class PredictiveMaintenanceSystem:
    """Main system managing industrial machine health monitoring"""

    def __init__(self, system_name='Machine Health Monitoring System'):
        self.system_name = system_name
        self.machines = {}
        self.system_status = 'operational'  # operational, maintenance, offline
        self.daily_statistics = []
        self.all_alerts = []
        self.maintenance_schedule = []

    def add_machine(self, machine_id, machine_name, machine_type='pump'):
        """Add a machine to the monitoring system"""
        self.machines[machine_id] = MachineProfile(machine_id, machine_name,
machine_type)

    def simulate_monitoring_day(self, date=None, num_intervals=96):
        """
        Simulate a full day of machine monitoring with 15-minute intervals
        """
        if date is None:
            date = datetime.now().date()

        daily_data = []
```

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```
        total_alerts = 0
        critical_alerts = 0
        maintenance_required = 0

        for interval in range(num_intervals):
            quarter_hour = (interval * 15) / 60
            hour = int(quarter_hour)
            minutes = int((quarter_hour % 1) * 60)

            current_time = datetime.combine(date, datetime.min.time()) +
timedelta(hours=quarter_hour)

            for machine_id, machine in self.machines.items():
                # Generate realistic operating parameters based on time of day
                load_factor = self._generate_load_factor(hour)

                # Simulate fault development over time (for demonstration)
                machine.fault_development = min(1.0, machine.fault_development +
np.random.uniform(0, 0.01))

                # Process each sensor for the machine
                for sensor_id, sensor in machine.sensors.items():
                    if isinstance(sensor, VibrationSensor):
                        base_vib = self._get_base_vibration(machine.machine_type)
                        measured_vib = sensor.read_vibration(base_vib, load_factor,
machine.fault_development)
                        alert = machine.fault_engine.evaluate_vibration(machine_id,
measured_vib, current_time)
                        if alert:
                            total_alerts += 1
                            if alert['severity'] == 'critical':
                                critical_alerts += 1
                            self.all_alerts.append(alert)

                    elif isinstance(sensor, TemperatureSensor):
                        base_temp = self._get_base_temperature(machine.machine_type)
                        measured_temp = sensor.read_temperature(base_temp, load_factor,
machine.fault_development)
                        alert = machine.fault_engine.evaluate_temperature(machine_id,
measured_temp, current_time)
                        if alert:
                            total_alerts += 1
                            if alert['severity'] == 'critical':
                                critical_alerts += 1
                            self.all_alerts.append(alert)
```

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```
                elif isinstance(sensor, CurrentSensor):
                    base_current = self._get_base_current(machine.machine_type)
                    measured_current = sensor.read_current(base_current, load_factor,
machine.fault_development)
                    alert = machine.fault_engine.evaluate_current(machine_id,
measured_current, current_time)
                    if alert:
                        total_alerts += 1
                        if alert['severity'] == 'critical':
                            critical_alerts += 1
                        self.all_alerts.append(alert)

            # Evaluate machine health
            machine.evaluate_machine_health(current_time)
            if machine.maintenance_due:
                maintenance_required += 1

            daily_data.append({ 'date
                ': date,
                'time': f"{hour:02d}:{minutes:02d}",
                'hour': hour,
                'machine_id': machine_id,
                'machine_name': machine.machine_name,
                'machine_type': machine.machine_type,
                'machine_status': machine.machine_status,
                'health_score': machine.health_score,
                'maintenance_due': machine.maintenance_due,
                'active_alerts': len(machine.fault_engine.alerts)
            })

        self.daily_statistics.append({ 'dat
            e': date,
            'total_alerts': total_alerts,
            'critical_alerts': critical_alerts,
            'maintenance_required': maintenance_required,
            'num_machines': len(self.machines),
            'system_status': self.system_status
        })

        return pd.DataFrame(daily_data)

    def _generate_load_factor(self, hour):
        """Generate realistic machine load patterns"""
        if 6 <= hour < 18:  # Day shift
            return np.random.uniform(0.6, 0.95)
        elif 18 <= hour < 22:  # Evening shift
```

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```
            return np.random.uniform(0.4, 0.8)
        else:  # Night shift
            return np.random.uniform(0.1, 0.5)

    def _get_base_vibration(self, machine_type):
        """Get baseline vibration for machine type"""
        vibration_map = {
            'pump': np.random.normal(1.2, 0.2),
            'motor': np.random.normal(1.5, 0.3),
            'compressor': np.random.normal(1.8, 0.3),
            'conveyor': np.random.normal(0.8, 0.2)
        }
        return vibration_map.get(machine_type, 1.0)

    def _get_base_temperature(self, machine_type):
        """Get baseline temperature for machine type"""
        temp_map = {
            'pump': np.random.normal(45, 3),
            'motor': np.random.normal(50, 4),
            'compressor': np.random.normal(55, 5),
            'conveyor': np.random.normal(35, 2)
        }
        return temp_map.get(machine_type, 40)

    def _get_base_current(self, machine_type):
        """Get baseline current for machine type"""
        current_map = {
            'pump': np.random.normal(10, 1),
            'motor': np.random.normal(12, 1.5),
            'compressor': np.random.normal(14, 2),
            'conveyor': np.random.normal(8, 1)
        }
        return current_map.get(machine_type, 10)

    def get_system_report(self):
        """Generate comprehensive system report"""
        report = {
            'system_name': self.system_name,
            'total_machines': len(self.machines),
            'total_alerts': len(self.all_alerts),
            'critical_alerts': len([a for a in self.all_alerts if a['severity'] == 'critical']),
            'simulation_days': len(self.daily_statistics),
            'system_status': self.system_status,
            'avg_daily_alerts': np.mean([stat['total_alerts'] for stat in self.daily_statistics]) if
self.daily_statistics else 0
        }
```

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```
        return report


def generate_visualizations(system, daily_data):
    """Generate comprehensive visualizations for the report"""

    # 1. Machine Health Metrics
    fig, axes = plt.subplots(2, 2, figsize=(14, 10))
    fig.suptitle('Industrial Machine Health Monitoring - Performance Metrics',
                 fontsize=16, fontweight='bold')

    # Health score by hour
    health_by_hour = daily_data.groupby('hour')['health_score'].mean()
    axes[0, 0].plot(health_by_hour.index, health_by_hour.values, marker='o', linewidth=2,
                    markersize=6, color='#2ECC71', label='Health Score')
    axes[0, 0].axhline(y=80, color='orange', linestyle='--', linewidth=2, label='Degradation
Threshold')
    axes[0, 0].axhline(y=40, color='red', linestyle='--', linewidth=2, label='Critical
Threshold')
    axes[0, 0].fill_between(health_by_hour.index, health_by_hour.values, alpha=0.3,
color='#2ECC71')
    axes[0, 0].set_title('Machine Health Score by Hour', fontweight='bold')
    axes[0, 0].set_ylabel('Health Score (%)')
    axes[0, 0].set_xlabel('Hour of Day')
    axes[0, 0].set_ylim([0, 105])
    axes[0, 0].legend()
    axes[0, 0].grid(True, alpha=0.3)

    # Machine status distribution
    status_counts = daily_data['machine_status'].value_counts()
    colors_status = {'healthy': '#2ECC71', 'degraded': '#F39C12', 'critical': '#E74C3C'}
    colors = [colors_status.get(status, 'gray') for status in status_counts.index]

    axes[0, 1].bar(status_counts.index, status_counts.values, color=colors,
edgecolor='black')
    axes[0, 1].set_title('Machine Status Distribution', fontweight='bold')
    axes[0, 1].set_ylabel('Frequency')
    axes[0, 1].set_xlabel('Machine Status')
    for i, v in enumerate(status_counts.values):
        axes[0, 1].text(i, v + 20, str(v), ha='center', fontweight='bold')

    # Maintenance requirements by hour
    maintenance_by_hour = daily_data.groupby('hour')['maintenance_due'].sum()
    axes[1, 0].bar(maintenance_by_hour.index, maintenance_by_hour.values,
color='#E74C3C',
                   edgecolor='black', alpha=0.7)
```

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```
    axes[1, 0].set_title('Maintenance Requirements by Hour', fontweight='bold')
    axes[1, 0].set_ylabel('Machines Requiring Maintenance')
    axes[1, 0].set_xlabel('Hour of Day')
    axes[1, 0].grid(True, alpha=0.3, axis='y')

    # Active alerts by hour
    alerts_by_hour = daily_data.groupby('hour')['active_alerts'].mean()
    axes[1, 1].plot(alerts_by_hour.index, alerts_by_hour.values, marker='s', linewidth=2.5,
                markersize=7, color='#E74C3C', label='Active Alerts')
    axes[1, 1].fill_between(alerts_by_hour.index, alerts_by_hour.values, alpha=0.3,
color='#E74C3C')
    axes[1, 1].set_title('Average Active Alerts by Hour', fontweight='bold')
    axes[1, 1].set_ylabel('Number of Alerts')
    axes[1, 1].set_xlabel('Hour of Day')
    axes[1, 1].legend()
    axes[1, 1].grid(True, alpha=0.3)

    plt.tight_layout()
    plt.savefig('/home/ubuntu/machine_monitoring_project/fig_health_metrics.png',
dpi=300, bbox_inches='tight')
    print("Saved: fig_health_metrics.png")
    plt.close()

    # 2. Fault Detection Analysis
    fig, axes = plt.subplots(2, 2, figsize=(14, 10))
    fig.suptitle('Fault Detection and Alert Analysis', fontsize=16, fontweight='bold')

    # Alert types distribution
    alert_types = [a['sensor_type'] for a in system.all_alerts]
    alert_counts = pd.Series(alert_types).value_counts()
    colors_map = {'vibration': '#E74C3C', 'temperature': '#F39C12', 'current': '#3498DB'}
    colors = [colors_map.get(alert, 'gray') for alert in alert_counts.index]

    axes[0, 0].bar(alert_counts.index, alert_counts.values, color=colors, edgecolor='black')
    axes[0, 0].set_title('Alert Types Distribution', fontweight='bold')
    axes[0, 0].set_ylabel('Count')
    axes[0, 0].set_xlabel('Sensor Type')
    for i, v in enumerate(alert_counts.values):
        axes[0, 0].text(i, v + 10, str(v), ha='center', fontweight='bold')

    # Alerts by hour
    alerts_by_hour = daily_data.groupby('hour')['active_alerts'].sum()
    axes[0, 1].bar(alerts_by_hour.index, alerts_by_hour.values, color='#E74C3C',
edgecolor='black', alpha=0.7)
    axes[0, 1].set_title('Total Alerts by Hour', fontweight='bold')
    axes[0, 1].set_ylabel('Number of Alerts')
```

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```
    axes[0, 1].set_xlabel('Hour of Day')
    axes[0, 1].grid(True, alpha=0.3, axis='y')

    # Alert severity distribution
    alert_severities = [a['severity'] for a in system.all_alerts]
    severity_counts = pd.Series(alert_severities).value_counts()
    colors_severity = {'medium': '#F39C12', 'high': '#E67E22', 'critical': '#E74C3C'}
    colors = [colors_severity.get(sev, 'gray') for sev in severity_counts.index]

    axes[1, 0].pie(severity_counts.values, labels=severity_counts.index, autopct='%1.1f%
%',
                   colors=colors, startangle=90, textprops={'fontsize': 11, 'fontweight': 'bold'})
    axes[1, 0].set_title('Alert Severity Distribution', fontweight='bold')

    # Machine-wise alert count
    machine_alerts = daily_data.groupby('machine_name')['active_alerts'].sum()
    axes[1, 1].barh(machine_alerts.index, machine_alerts.values, color='#E74C3C',
edgecolor='black')
    axes[1, 1].set_title('Total Alerts by Machine', fontweight='bold')
    axes[1, 1].set_xlabel('Number of Alerts')
    for i, v in enumerate(machine_alerts.values):
        axes[1, 1].text(v + 5, i, str(int(v)), va='center', fontweight='bold')

    plt.tight_layout()
    plt.savefig('/home/ubuntu/machine_monitoring_project/fig_fault_analysis.png',
dpi=300, bbox_inches='tight')
    print("Saved: fig_fault_analysis.png")
    plt.close()

    # 3. System Architecture Diagram
    fig, ax = plt.subplots(figsize=(12, 8))
    ax.axis('off')

    ax.text(0.5, 0.95, 'Industrial IoT Machine Health Monitoring System Architecture',
            ha='center', fontsize=16, fontweight='bold', transform=ax.transAxes)

    components =
        [ ('Vibration\nSensor', 0.15,
        0.75),
        ('Temperature\nSensor', 0.5, 0.75),
        ('Current\nSensor', 0.85, 0.75),
        ('IoT\nCommunication\nModule', 0.5, 0.55),
        ('Cloud\nServer', 0.15, 0.35),
        ('Fault\nDetection', 0.5, 0.35),
        ('Maintenance\nScheduler', 0.85, 0.35),
    ]
```

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```
    for label, x, y in components:
        bbox = dict(boxstyle='round,pad=0.6', facecolor='lightblue', edgecolor='black',
linewidth=2)
        ax.text(x, y, label, ha='center', va='center', fontsize=11, fontweight='bold',
                transform=ax.transAxes, bbox=bbox)

    connections = [
        ((0.25, 0.75), (0.4, 0.75)),
        ((0.6, 0.75), (0.75, 0.75)),
        ((0.5, 0.65), (0.5, 0.60)),
        ((0.25, 0.65), (0.25, 0.60)),
        ((0.75, 0.65), (0.75, 0.60)),
    ]

    for start, end in connections:
        ax.annotate('', xy=end, xytext=start, transform=ax.transAxes,
                    arrowprops=dict(arrowstyle='->', lw=2,
                    color='black'))

    features_text = 'Key Features:\n• Real-time Machine Condition Monitoring\n•
Multi-sensor Data Fusion\n• Predictive Fault Detection\n• Automated Maintenance
Scheduling\n• Industry 4.0 Compliance'
    ax.text(0.5, 0.15, features_text, ha='center', va='top', fontsize=10,
            transform=ax.transAxes, bbox=dict(boxstyle='round', facecolor='lightyellow',
            edgecolor='black', linewidth=1.5))

    plt.tight_layout()
    plt.savefig('/home/ubuntu/machine_monitoring_project/fig_system_architecture.png',
dpi=300, bbox_inches='tight')
    print("Saved: fig_system_architecture.png")
    plt.close()

    # 4. Daily Performance
    fig, ax = plt.subplots(figsize=(12, 6))

    daily_stats = pd.DataFrame(system.daily_statistics)

    x = np.arange(len(daily_stats))
    width = 0.35

    ax.bar(x - width/2, daily_stats['total_alerts'], width, label='Total Alerts',
           color='#F39C12', edgecolor='black')
    ax2 = ax.twinx()
    ax2.bar(x + width/2, daily_stats['critical_alerts'], width, label='Critical Alerts',
            color='#E74C3C', alpha=0.7, edgecolor='black')

    ax.set_xlabel('Day', fontweight='bold')
```

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```
    ax.set_ylabel('Total Alerts', fontweight='bold', color='#F39C12')
    ax2.set_ylabel('Critical Alerts', fontweight='bold', color='#E74C3C')
    ax.set_title('Daily System Performance and Alert Trends', fontweight='bold',
fontsize=14)
    ax.set_xticks(x)
    ax.set_xticklabels([f"Day {i+1}" for i in range(len(daily_stats))])
    ax.grid(True, alpha=0.3, axis='y')

    lines1, labels1 = ax.get_legend_handles_labels()
    lines2, labels2 = ax2.get_legend_handles_labels()
    ax.legend(lines1 + lines2, labels1 + labels2, loc='upper left')

    plt.tight_layout()
    plt.savefig('/home/ubuntu/machine_monitoring_project/fig_daily_performance.png',
dpi=300, bbox_inches='tight')
    print("Saved: fig_daily_performance.png")
    plt.close()

    # 5. Machine Condition Heatmap
    fig, ax = plt.subplots(figsize=(12,
    6))

    machine_activity = daily_data.pivot_table(values='health_score',
index='machine_name',
                                              columns='hour', aggfunc='mean', fill_value=0)

    sns.heatmap(machine_activity, cmap='RdYlGn', annot=True, fmt='.0f',
cbar_kws={'label': 'Health Score'},
                ax=ax, linewidths=0.5, linecolor='gray', vmin=0, vmax=100)
    ax.set_title('Machine Health Score Heatmap - Health by Machine and Hour',
fontweight='bold', fontsize=14)
    ax.set_xlabel('Hour of Day', fontweight='bold')
    ax.set_ylabel('Machine Name', fontweight='bold')

    plt.tight_layout()
    plt.savefig('/home/ubuntu/machine_monitoring_project/fig_health_heatmap.png',
dpi=300, bbox_inches='tight')
    print("Saved: fig_health_heatmap.png")
    plt.close()


def main():
    """Main execution function"""
    print("=" * 70)
    print("INDUSTRIAL IOT-BASED MACHINE HEALTH MONITORING SYSTEM
SIMULATION")
```

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```
print("=" * 70)

# Initialize system
system = PredictiveMaintenanceSystem("Industrial Machine Health Monitoring
System")

# Add machines with different types
machines_config = {
    'MACH-001': {'name': 'Centrifugal Pump A', 'type': 'pump'},
    'MACH-002': {'name': 'Electric Motor B', 'type': 'motor'},
    'MACH-003': {'name': 'Air Compressor C', 'type': 'compressor'},
    'MACH-004': {'name': 'Conveyor System D', 'type': 'conveyor'},
    'MACH-005': {'name': 'Rotary Pump E', 'type': 'pump'},
}

for machine_id, config in machines_config.items():
    system.add_machine(machine_id, config['name'], config['type'])
    machine = system.machines[machine_id]

    # Add sensors to each machine
    machine.add_sensor(f"{machine_id}-VIB-X", 'vibration',
    axis='X') machine.add_sensor(f"{machine_id}-VIB-Y", 'vibration',
    axis='Y')
    machine.add_sensor(f"{machine_id}-TEMP-B", 'temperature',
    location='bearing') machine.add_sensor(f"{machine_id}-CURR-A", 'current',
    phase='A')

print(f"\nSystem initialized with {len(machines_config)} machines:")
for machine_id, config in machines_config.items():
    print(f"  - {machine_id}: {config['name']} ({config['type']})")

# Simulate 7 days
print("\nSimulating 7 days of machine monitoring...")
all_data = []

for day in range(7):
    simulation_date = datetime.now().date() - timedelta(days=6-day)
    daily_data = system.simulate_monitoring_day(simulation_date)
    all_data.append(daily_data)

    daily_stats = system.daily_statistics[-1]
    print(f"Day {day+1}: Alerts={daily_stats['total_alerts']}, "
          f"Critical={daily_stats['critical_alerts']}, "
          f"Maintenance={daily_stats['maintenance_required']}")

# Combine all data
combined_data = pd.concat(all_data, ignore_index=True)
```

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```
    # Generate visualizations
    print("\nGenerating visualizations...")
    generate_visualizations(system, combined_data)

    # Generate report data
    report = system.get_system_report()

    print("\n" + "=" * 70)
    print("SYSTEM REPORT")
    print("=" * 70)
    print(f"System Name: {report['system_name']}")
    print(f"Total Machines Monitored: {report['total_machines']}")
    print(f"Total Alerts Generated: {report['total_alerts']}")
    print(f"Critical Alerts: {report['critical_alerts']}")
    print(f"Average Daily Alerts: {report['avg_daily_alerts']:.1f}")
    print(f"System Status: {report['system_status']}")
    print(f"Simulation Period: {report['simulation_days']} days")
    print("=" * 70)

    # Save data to CSV

combined_data.to_csv('/home/ubuntu/machine_monitoring_project/monitoring_data.csv',
index=False)
    print("\nMonitoring data saved to: monitoring_data.csv")

    # Save report to JSON
    report_json = {
        'system_report': report,
        'machines': list(machines_config.keys()),
        'simulation_date': datetime.now().isoformat(),
        'daily_statistics': system.daily_statistics,
        'total_alerts': len(system.all_alerts)
    }

    with open('/home/ubuntu/machine_monitoring_project/system_report.json', 'w') as f:
        json.dump(report_json, f, indent=2, default=str)

    print("System report saved to: system_report.json")

    return system, combined_data


if __name__ == "__main__":
    system, data = main()
```

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CHAPTER 9

APPENDIX B: SIMULATION DATA SAMPLE

This appendix contains a sample of the raw simulation data generated by the system, demonstrating the granular 15-minute interval logging of machine operational parameters.

Time	Machine ID	Status	Health Score	Maint. Due	Alerts
00:00	MACH-001	healthy	100.0	False	0
00:00	MACH-002	degraded	75.0	False	1
00:00	MACH-003	healthy	100.0	False	0
00:00	MACH-004	healthy	100.0	False	0
00:00	MACH-005	healthy	100.0	False	0
00:15	MACH-001	healthy	100.0	False	0
00:15	MACH-002	healthy	100.0	False	1
00:15	MACH-003	degraded	75.0	False	1
00:15	MACH-004	healthy	100.0	False	0
00:15	MACH-005	healthy	100.0	False	0
00:30	MACH-001	healthy	100.0	False	0
00:30	MACH-002	healthy	100.0	False	1
00:30	MACH-003	degraded	75.0	False	2

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00:30	MACH-004	healthy	100.0	False	0
00:30	MACH-005	healthy	100.0	False	0
00:45	MACH-001	healthy	100.0	False	0
00:45	MACH-002	healthy	100.0	False	1
00:45	MACH-003	degraded	70.0	False	4
00:45	MACH-004	healthy	100.0	False	0
00:45	MACH-005	healthy	100.0	False	0
01:00	MACH-001	healthy	100.0	False	0
01:00	MACH-002	healthy	100.0	False	1
01:00	MACH-003	healthy	100.0	False	4
01:00	MACH-004	healthy	100.0	False	0
01:00	MACH-005	healthy	100.0	False	0
01:15	MACH-001	healthy	100.0	False	0
01:15	MACH-002	healthy	100.0	False	1
01:15	MACH-003	degraded	75.0	False	5
01:15	MACH-004	healthy	100.0	False	0
01:15	MACH-005	healthy	100.0	False	0
01:30	MACH-001	healthy	100.0	False	0
01:30	MACH-002	healthy	100.0	False	1
01:30	MACH-003	degraded	70.0	False	7

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01:30	MACH-004	healthy	100.0	False	0
01:30	MACH-005	healthy	100.0	False	0
01:45	MACH-001	healthy	100.0	False	0
01:45	MACH-002	healthy	100.0	False	1
01:45	MACH-003	healthy	100.0	False	7
01:45	MACH-004	healthy	100.0	False	0
01:45	MACH-005	healthy	100.0	False	0
02:00	MACH-001	healthy	100.0	False	0
02:00	MACH-002	degraded	75.0	False	2
02:00	MACH-003	degraded	70.0	False	9
02:00	MACH-004	healthy	100.0	False	0
02:00	MACH-005	healthy	100.0	False	0
02:15	MACH-001	healthy	100.0	False	0
02:15	MACH-002	healthy	100.0	False	2
02:15	MACH-003	degraded	70.0	False	11
02:15	MACH-004	healthy	100.0	False	0
02:15	MACH-005	healthy	100.0	False	0
02:30	MACH-001	degraded	75.0	False	1
02:30	MACH-002	healthy	100.0	False	2

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02:30	MACH-003	degraded	70.0	False	13
02:30	MACH-004	healthy	100.0	False	0
02:30	MACH-005	healthy	100.0	False	0
02:45	MACH-001	healthy	100.0	False	1
02:45	MACH-002	degraded	75.0	False	3
02:45	MACH-003	degraded	75.0	False	14
02:45	MACH-004	healthy	100.0	False	0
02:45	MACH-005	healthy	100.0	False	0
03:00	MACH-001	healthy	100.0	False	1
03:00	MACH-002	healthy	100.0	False	3
03:00	MACH-003	degraded	70.0	False	16
03:00	MACH-004	healthy	100.0	False	0
03:00	MACH-005	healthy	100.0	False	0
03:15	MACH-001	healthy	100.0	False	1
03:15	MACH-002	healthy	100.0	False	3
03:15	MACH-003	degraded	70.0	False	18
03:15	MACH-004	healthy	100.0	False	0
03:15	MACH-005	healthy	100.0	False	0
03:30	MACH-001	healthy	100.0	False	1

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03:30	MACH-002	healthy	100.0	False	3
03:30	MACH-003	degraded	75.0	False	19
03:30	MACH-004	healthy	100.0	False	0
03:30	MACH-005	healthy	100.0	False	0
03:45	MACH-001	healthy	100.0	False	1
03:45	MACH-002	degraded	75.0	False	4
03:45	MACH-003	degraded	75.0	False	20
03:45	MACH-004	healthy	100.0	False	0
03:45	MACH-005	healthy	100.0	False	0
04:00	MACH-001	healthy	100.0	False	1
04:00	MACH-002	healthy	100.0	False	4
04:00	MACH-003	degraded	75.0	False	21
04:00	MACH-004	healthy	100.0	False	0
04:00	MACH-005	healthy	100.0	False	0
04:15	MACH-001	healthy	100.0	False	1
04:15	MACH-002	healthy	100.0	False	4
04:15	MACH-003	degraded	75.0	False	22
04:15	MACH-004	healthy	100.0	False	0
04:15	MACH-005	healthy	100.0	False	0

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04:30	MACH-001	healthy	100.0	False	1
04:30	MACH-002	degraded	75.0	False	5
04:30	MACH-003	degraded	70.0	False	24
04:30	MACH-004	healthy	100.0	False	0
04:30	MACH-005	healthy	100.0	False	0
04:45	MACH-001	healthy	100.0	False	1
04:45	MACH-002	degraded	75.0	False	6
04:45	MACH-003	degraded	75.0	False	25
04:45	MACH-004	healthy	100.0	False	0
04:45	MACH-005	healthy	100.0	False	0

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CHAPTER 10

APPENDIX C: EXTENDED LITERATURE REVIEW AND METHODOLOGY

Extended discussion on Industrial IoT Sensor Calibration 1. The deployment of non-invasive vibration sensors in predictive maintenance requires rigorous calibration protocols to ensure operational accuracy. Piezoelectric accelerometers used for high-frequency vibration monitoring, while accurate, are sensitive to mounting resonance and ambient machine noise interference. Implementing a dynamic signal processing algorithm (such as Fast Fourier Transform - FFT) that filters out background noise based on integrated operational load data is crucial for minimizing false maintenance alarms. Furthermore, sensor degradation and environmental exposure over time necessitate a robust device management model. By continuously monitoring signal-to-noise ratios, the system can alert technicians to recalibrate the sensor or replace the device, ensuring uninterrupted operational observation.

Detailed analysis of Industrial Communication Protocols and Security 1. In the event of a critical equipment anomaly, the speed, reliability, and security of the alert transmission are paramount. Industrial data transmission must comply with strict operational standards and cybersecurity frameworks. Utilizing end-to-end encrypted OPC UA or MQTT over TLS ensures that machine telemetry is delivered securely to maintenance dashboards within milliseconds. To guarantee delivery even in adverse network conditions typical of heavy manufacturing environments, the system architecture must include edge computing capabilities on the local PLC or gateway device to process critical alerts locally and trigger automated shutdown sequences, while utilizing secondary industrial networks as a fallback if the primary Wi-Fi connection is disrupted.

Extended discussion on Industrial IoT Sensor Calibration 2. The deployment of non-invasive vibration sensors in predictive maintenance requires rigorous calibration protocols to ensure operational accuracy. Piezoelectric accelerometers used for high-frequency vibration monitoring, while accurate, are sensitive to mounting resonance and

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Extended discussion on Industrial IoT Sensor Calibration 3. The deployment of non-invasive vibration sensors in predictive maintenance requires rigorous calibration protocols to ensure operational accuracy. Piezoelectric accelerometers used for high-frequency vibration monitoring, while accurate, are sensitive to mounting resonance and ambient machine noise interference. Implementing a dynamic signal processing algorithm (such as Fast Fourier Transform - FFT) that filters out background noise based on integrated operational load data is crucial for minimizing false maintenance alarms. Furthermore, sensor degradation and environmental exposure over time necessitate a robust device management model. By continuously monitoring signal-to-noise ratios, the system can alert technicians to recalibrate the sensor or replace the device, ensuring uninterrupted operational observation.

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Extended discussion on Industrial IoT Sensor Calibration 4. The deployment of non-invasive vibration sensors in predictive maintenance requires rigorous calibration protocols to ensure operational accuracy. Piezoelectric accelerometers used for high-frequency vibration monitoring, while accurate, are sensitive to mounting resonance and ambient machine noise interference. Implementing a dynamic signal processing algorithm (such as Fast Fourier Transform - FFT) that filters out background noise based on integrated operational load data is crucial for minimizing false maintenance alarms. Furthermore, sensor degradation and environmental exposure over time necessitate a robust device management model. By continuously monitoring signal-to-noise ratios, the system can alert technicians to recalibrate the sensor or replace the device, ensuring uninterrupted operational observation.

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Extended discussion on Industrial IoT Sensor Calibration 6. The deployment of non-invasive vibration sensors in predictive maintenance requires rigorous calibration protocols to ensure operational accuracy. Piezoelectric accelerometers used for high-frequency vibration monitoring, while accurate, are sensitive to mounting resonance and ambient machine noise interference. Implementing a dynamic signal processing algorithm (such as Fast Fourier Transform - FFT) that filters out background noise based

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on integrated operational load data is crucial for minimizing false maintenance alarms. Furthermore, sensor degradation and environmental exposure over time necessitate a robust device management model. By continuously monitoring signal-to-noise ratios, the system can alert technicians to recalibrate the sensor or replace the device, ensuring uninterrupted operational observation.

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Extended discussion on Industrial IoT Sensor Calibration 7. The deployment of non-invasive vibration sensors in predictive maintenance requires rigorous calibration protocols to ensure operational accuracy. Piezoelectric accelerometers used for high-frequency vibration monitoring, while accurate, are sensitive to mounting resonance and ambient machine noise interference. Implementing a dynamic signal processing algorithm (such as Fast Fourier Transform - FFT) that filters out background noise based on integrated operational load data is crucial for minimizing false maintenance alarms. Furthermore, sensor degradation and environmental exposure over time necessitate a robust device management model. By continuously monitoring signal-to-noise ratios, the system can alert technicians to recalibrate the sensor or replace the device, ensuring uninterrupted operational observation.

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Extended discussion on Industrial IoT Sensor Calibration 9. The deployment of non-invasive vibration sensors in predictive maintenance requires rigorous calibration protocols to ensure operational accuracy. Piezoelectric accelerometers used for high-frequency vibration monitoring, while accurate, are sensitive to mounting resonance and ambient machine noise interference. Implementing a dynamic signal processing algorithm (such as Fast Fourier Transform - FFT) that filters out background noise based on integrated operational load data is crucial for minimizing false maintenance alarms. Furthermore, sensor degradation and environmental exposure over time necessitate a robust device management model. By continuously monitoring signal-to-noise ratios, the system can alert technicians to recalibrate the sensor or replace the device, ensuring uninterrupted operational observation.

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Extended discussion on Industrial IoT Sensor Calibration 10. The deployment of non-invasive vibration sensors in predictive maintenance requires rigorous calibration protocols to ensure operational accuracy. Piezoelectric accelerometers used for high-frequency vibration monitoring, while accurate, are sensitive to mounting resonance and ambient machine noise interference. Implementing a dynamic signal processing algorithm (such as Fast Fourier Transform - FFT) that filters out background noise based on integrated operational load data is crucial for minimizing false maintenance alarms. Furthermore, sensor degradation and environmental exposure over time necessitate a robust device management model. By continuously monitoring signal-to-noise ratios, the

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Extended discussion on Industrial IoT Sensor Calibration 11. The deployment of non-invasive vibration sensors in predictive maintenance requires rigorous calibration protocols to ensure operational accuracy. Piezoelectric accelerometers used for high-frequency vibration monitoring, while accurate, are sensitive to mounting resonance and ambient machine noise interference. Implementing a dynamic signal processing algorithm (such as Fast Fourier Transform - FFT) that filters out background noise based on integrated operational load data is crucial for minimizing false maintenance alarms. Furthermore, sensor degradation and environmental exposure over time necessitate a robust device management model. By continuously monitoring signal-to-noise ratios, the system can alert technicians to recalibrate the sensor or replace the device, ensuring uninterrupted operational observation.

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Extended discussion on Industrial IoT Sensor Calibration 12. The deployment of non-invasive vibration sensors in predictive maintenance requires rigorous calibration protocols to ensure operational accuracy. Piezoelectric accelerometers used for high-frequency vibration monitoring, while accurate, are sensitive to mounting resonance and ambient machine noise interference. Implementing a dynamic signal processing algorithm (such as Fast Fourier Transform - FFT) that filters out background noise based on integrated operational load data is crucial for minimizing false maintenance alarms. Furthermore, sensor degradation and environmental exposure over time necessitate a robust device management model. By continuously monitoring signal-to-noise ratios, the system can alert technicians to recalibrate the sensor or replace the device, ensuring uninterrupted operational observation.

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Extended discussion on Industrial IoT Sensor Calibration 13. The deployment of non-invasive vibration sensors in predictive maintenance requires rigorous calibration protocols to ensure operational accuracy. Piezoelectric accelerometers used for high-frequency vibration monitoring, while accurate, are sensitive to mounting resonance and

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Extended discussion on Industrial IoT Sensor Calibration 14. The deployment of non-invasive vibration sensors in predictive maintenance requires rigorous calibration protocols to ensure operational accuracy. Piezoelectric accelerometers used for high-frequency vibration monitoring, while accurate, are sensitive to mounting resonance and ambient machine noise interference. Implementing a dynamic signal processing algorithm (such as Fast Fourier Transform - FFT) that filters out background noise based on integrated operational load data is crucial for minimizing false maintenance alarms. Furthermore, sensor degradation and environmental exposure over time necessitate a robust device management model. By continuously monitoring signal-to-noise ratios, the system can alert technicians to recalibrate the sensor or replace the device, ensuring uninterrupted operational observation.

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Detailed analysis of Industrial Communication Protocols and Security 14. In the event of a critical equipment anomaly, the speed, reliability, and security of the alert transmission are paramount. Industrial data transmission must comply with strict operational standards and cybersecurity frameworks. Utilizing end-to-end encrypted OPC UA or MQTT over TLS ensures that machine telemetry is delivered securely to maintenance dashboards within milliseconds. To guarantee delivery even in adverse network conditions typical of heavy manufacturing environments, the system architecture must include edge computing capabilities on the local PLC or gateway device to process critical alerts locally and trigger automated shutdown sequences, while utilizing secondary industrial networks as a fallback if the primary Wi-Fi connection is disrupted.

Extended discussion on Industrial IoT Sensor Calibration 15. The deployment of non-invasive vibration sensors in predictive maintenance requires rigorous calibration protocols to ensure operational accuracy. Piezoelectric accelerometers used for high-frequency vibration monitoring, while accurate, are sensitive to mounting resonance and ambient machine noise interference. Implementing a dynamic signal processing algorithm (such as Fast Fourier Transform - FFT) that filters out background noise based on integrated operational load data is crucial for minimizing false maintenance alarms. Furthermore, sensor degradation and environmental exposure over time necessitate a robust device management model. By continuously monitoring signal-to-noise ratios, the system can alert technicians to recalibrate the sensor or replace the device, ensuring uninterrupted operational observation.

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Detailed analysis of Industrial Communication Protocols and Security 15. In the event of a critical equipment anomaly, the speed, reliability, and security of the alert transmission are paramount. Industrial data transmission must comply with strict operational standards and cybersecurity frameworks. Utilizing end-to-end encrypted OPC UA or MQTT over TLS ensures that machine telemetry is delivered securely to maintenance dashboards within milliseconds.